Graph-Weighted Canonical Correlation Analysis

A Closed-Form Method for Inductive Attributed Graph Embedding

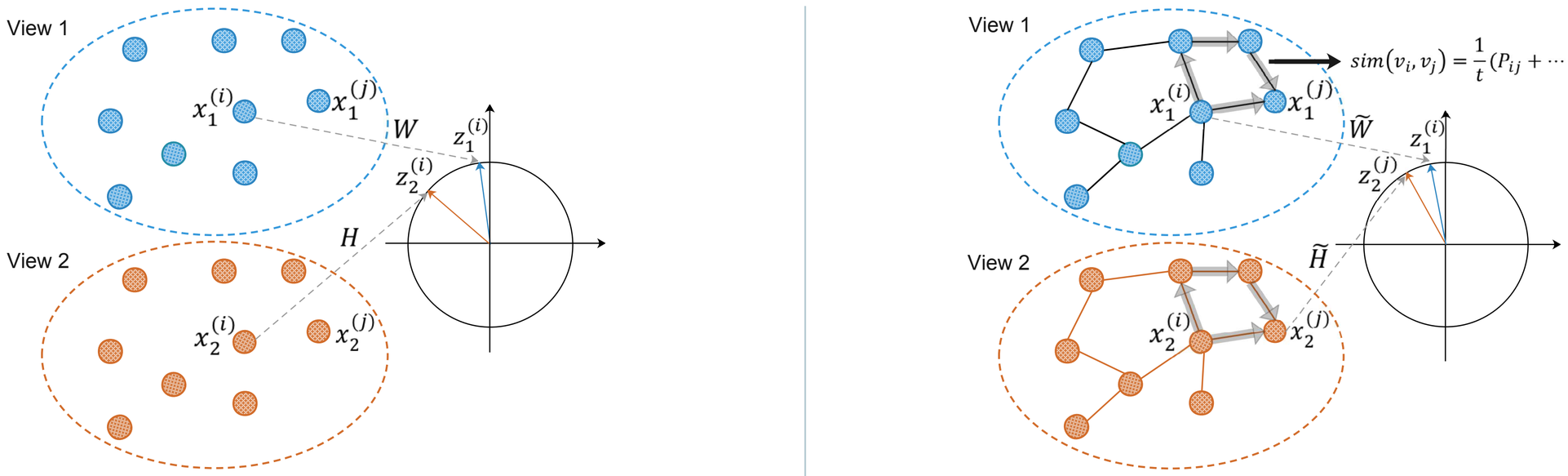
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LoG 2026 · PMLR Volume 338 · github.com/jianjinuta/gw-cca

CORE IDEA · FIGURE 1

Replace same-index alignment with graph-weighted alignment



(a) Standard CCA

aligns only same-index pairs (*i*, *i*)

$$\max \operatorname{Tr}(W^\top X_1^\top X_2 H)$$

(b) GW-CCA

aligns every pair (*i*, *j*) in proportion to graph similarity M_{ij}

$$\max \operatorname{Tr}(\widetilde{W}^\top X_1^\top M X_2 \widetilde{H})$$

Problem

Inductive graph embedding: embed nodes never seen during training from their attributes and neighbourhood, without retraining.

Question. Can a closed-form, deterministic method use graph structure well enough to compete with trained GNN and graph-SSL encoders?

Strict inductive protocol. The encoder is built on the training subgraph; test nodes and their edges are seen only at inference.

- **5 benchmarks:** ogbn-arxiv, Reddit, Yelp, ogbn-products, PPI
- **10 baselines:** supervised GNN, contrastive, masked-AE and linear
- **Ablation CCA*:** the same two views with $M = I$

Closed-form method

Two deterministic views of the node attributes **X**:

$$X_1 = X, \qquad X_2 = \widehat{P}^K X$$
$$M = \tfrac{1}{2}(P + P^2), \qquad P = D^{-1} A$$

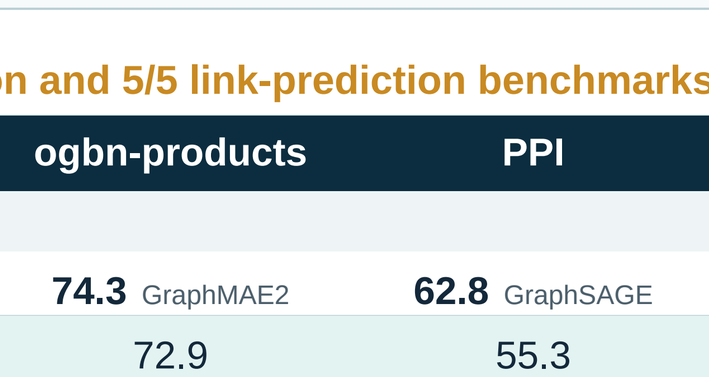
1. **Propagate.** *K* sparse products give X_2; no dense graph powers.
2. **Whiten.** Cholesky-factor the ridged $p \times p$ covariances.
3. **Solve once.** One truncated rank-*d* SVD:
$$\mathrm{SVD}_d\left(C_{11}^{-1/2} X_1^\top M X_2 C_{22}^{-1/2}\right)$$
4. **Infer.** Unseen nodes are embedded by fixed projections:
$$Z' = [\, X'\widetilde{W} \,\|\, \widehat{P}'^K X'\widetilde{H} \,]$$

Cost no GPU · no seeds · no augmentations

O(K|E|p + np² + p³)

What does M buy?

Gain of GW-CCA over CCA* (same views, $M = I$), percentage points:



Largest gains on pairwise link prediction, where *M* weights exactly the node pairs the task scores.

Limitation. Trails GraphSAGE on PPI node classification (55.3 vs 62.8): its small graphs favour deep message passing over shallow propagation.

Benchmark results

Best on 3/5 node-classification and 5/5 link-prediction benchmarks

Method	ogbn-arxiv	Reddit	Yelp	ogbn-products	PPI
Node classification accuracy / Micro-F1 (%)					
Strongest baseline	69.5 SGC	94.6 GraphMAE2	57.1 GraphMAE2	**74.3** GraphMAE2	**62.8** GraphSAGE
GW-CCA	**69.7**	**96.3**	**61.1**	72.9	55.3
Link prediction AUC (%)					
Strongest baseline	95.3 GraphACL	94.9 GraphSAGE	93.0 GRACE	98.1 GGD	87.5 GRACE
GW-CCA	**97.5**	**97.6**	**93.7**	**99.5**	**89.2**
Encoder construction time seconds; GW-CCA on CPU, deep baselines on one A10G GPU					
Fastest deep baseline	11 CCA-SSG	125 GraphACL	226 GGD	139 GraphACL	11 CCA-SSG
GW-CCA (CPU)	**1.8**	**42.0**	**28.1**	**12.1**	**0.3**

Strongest baseline = best of the 10 baselines per column, excluding the CCA* ablation (deep baselines: mean over 5 seeds; GW-CCA is deterministic). Bold = best in column. Timings are end-to-end but not hardware-matched (CPU vs GPU). Full tables with standard deviations: paper Tables 1–3.

TAKEAWAY

Graph-weighting the cross-covariance turns CCA into a competitive inductive graph encoder: one closed-form SVD, no iterative training.

Code: github.com/jianjinuta/gw-cca

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